

ANUPREET SINGH

anupreet2226579@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#) | +1 (410) 260-9818

EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript, C++, SQL

Tools and Frameworks: REST(Django, FastAPI, Next.js), React, PostgreSQL, AWS, Redis, Kafka, Docker, Kubernetes, GitHub Actions(CI/CD), Git, Linux, PyTest, Jest, PyTorch, LangChain

Areas of Interest and Coursework: Distributed System Design, AI & LLMs, Software Engineering (SDLC)

INDUSTRY EXPERIENCE

Data AI Engineer | *Platinum Business Services (R&D Team)* | Catonsville, MD

Dec 2025 – Present

- Developed **backend services** for an AI-native GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.
- Built a priority-ordered fallback in the **RAG chat agent** across 5 embedding models (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising evidence-backed control assessment accuracy from **64% to 92%**.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4-5 GB model weights lingering in the OS page cache, and fixed it with targeted `posix_fadvise` eviction, eliminating model-load failures.

Software Engineer (AI) Intern | *PMA* | Tallahassee, FL

Sep 2025 – Dec 2025

- Built an **AI-powered** PRD review platform in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.
- Architected a fully **serverless** review pipeline on AWS (S3 + CloudFront-hosted React SPA, Django on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining **200+** submissions across 5 cohorts at ~\$3/month idle cost.

Software Development Engineer | *Turfco Landscape Co.* | Surrey, BC

Jan 2023 – June 2024

- Architected a self-serve quoting platform that priced jobs with deterministic multi-stage evaluation, JSONB encoded conditions, per-quote audit trails of every rule applied, and a write-invalidated **Redis** cache, all driven by rules editable from an admin console instead of a developer deploy, cutting quote turnaround from ~2 days to <1 min.
- Built the online ordering and booking platform by taking self-serve orders through Stripe Checkout deposits, confirming them with signature-verified idempotent webhooks, and relaying a transactional outbox to **Kafka** workers for order confirmation, customer notification, and crew allocation, taking **\$600K+** in online bookings across ~1,100 orders.
- Delivered the **React SPA** that carried a customer from property details to instant estimate to paid deposit in one client-routed flow, with role-gated admin routes and JWT access tokens backed by httpOnly refresh cookies, enabling a streamlined booking flow with ~40% of bookings occurring outside office hours without phone support.
- Designed and deployed the **AWS architecture** that took the business off a WordPress brochure site, with SPA on CloudFront, autoscaling Express tier behind an ALB across two AZs, Multi-AZ RDS, sized for ~10x seasonal peaks and load-tested to ~50 quote requests/sec at <120 ms p99 with zero-downtime rolling deploys.

ACADEMIC EXPERIENCE

Graduate Assistant | *UMBC - Data Structures & Software Engineering*

Jan 2025 – June 2026

- Designed over 10 C++ assignments on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for 40+ students.
- Developed 200+ unit tests for the DSA projects on Linked Lists, Self Balancing Trees, Heaps, and Hash Tables automating logic checking and reducing grading time by over 80%.
- Helped 50+ students design and debug C++ projects implementing core data structures, and graded their homework.
- Held office hours to clarify technical concepts and translate requirements into actionable implementation steps.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.